KAWALTUMBUH: AN INTEGRATED OFFLINE-FIRST MOBILE APPLICATION FOR GROWTH MONITORING, NUTRITION GUIDANCE, AND IMMUNIZATION TRACKING TO SUPPORT STUNTING PREVENTION

COMPILED BY

Muhammad Albharaka Putrosandy
Maya Sirrurrifka
Salma Mardhiyah
Muhammad Fathir
Raden Yasmeen Rahima
Sheryl Talitha Uma

UNIVERSITAS SYIAH KUALA
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ABSTRACT

Stunting is a chronic nutrition problem that still affects 19.8% of children under five in Indonesia (about 4.48 million children) and contributes to roughly one million child deaths worldwide every year. Early detection at the household level remains difficult because of low nutrition literacy, Posyandu (community health post) check-ups that take place only once a month, MPASI (complementary feeding) menus that do not take family budgets into account, and immunization schedules that are easily missed. This study aims to develop KawalTumbuh, an offline-first Android application that integrates Z-score-based child growth monitoring, budget-friendly MPASI recipe recommendations using local ingredients, immunization schedule tracking, and pregnancy nutrition guidance. The application was developed using the prototyping model and runs entirely on the device without a backend, so it keeps working without an internet connection and children's data never leaves the device. Nutritional status is calculated using the WHO standard LMS method and classified according to Permenkes (Minister of Health Regulation) No. 2 of 2020, after which energy and macronutrient targets are adjusted to the child's status to rank recipes. Testing included black-box testing, verification of Z-scores against WHO tables, and a usability test using a Likert-scale questionnaire with 30 parents of children under five. The black-box test ran eight scenarios, each simulating a real user flow in the application from first-time setup to everyday use. All eight scenarios passed, and the maximum Z-score difference from the WHO reference was only 0.01 SD, well below the 0.1 SD tolerance. The usability test produced an average score of 4.65 (93.00%), in the Very Good category. KawalTumbuh has the potential to become a self-help tool for parents in supporting stunting prevention.

Keywords: stunting; Z-score; MPASI; immunization; offline-first Android application

CHAPTER 1

Introduction

Although the national trend has begun to improve, stunting remains one of Indonesia's main chronic nutrition emergencies. According to the 2024 Indonesian Nutritional Status Survey (SSGI), stunting prevalence was reduced to 19.8 percent (4,482,340 children under five), down 1.7 percent from the previous year. But unfortunately, this decline has not been even across the country. Eleven out of 38 provinces still recorded an increase in cases, with the highest stunting rate reaching 29.8 percent in the poorest economic group [1].

One of the root causes at the household level is low nutrition literacy and late detection, as stunting is often only noticed after it has been going on for a long time [2]. This problem is made worse by several practical obstacles in the field. Most people still find it hard to understand the nutritional content of the food they eat every day because nutrition information feels complicated and difficult to understand [3]. Meanwhile, parents often only learn about their child's nutritional status after a check-up at the Posyandu, which is held only once a month [6]. The MPASI menu recommendations in circulation also often do not consider limited family budgets, even though nutritious food does not have to be expensive and can rely on affordable local protein sources such as eggs and fish [4]. In addition, children's immunization schedules are often because parents lack an accessible, automatic reminder they can use at home [5].

These various limitations of conventional monitoring open up an opportunity to use digital technology as a more independent alternative solution. Various studies on developing Z-score-based digital applications, such as StunCare [6] and the Stunting Detection Calculator/"Kalkulating" [2], have shown that digital tools can calculate nutritional status accurately and speed up risk identification. However, most existing applications stop at Z-score calculation and have not yet brought daily nutrition, economic practicality, and preventive health monitoring such as immunization together into a single ecosystem.

Building on this need, KawalTumbuh was designed as an integrated platform with four main features: a Z-score-based child nutrition calculator for early detection of growth status, budget-adjusted MPASI recipe recommendations using local ingredients, immunization schedule tracking to support children's preventive health, and pregnancy nutrition guidance for expectant mothers. KawalTumbuh is expected to serve as a self-help intervention tool at the household level to support the government's target of reducing national stunting prevalence to 14.2 percent by 2029, while also making a real contribution to the global agenda of SDG 2.2 and SDG 3 [7].

CHAPTER 2
Literature Review

2.1 Stunting

2.1.1 Definition of Stunting

Stunting is a chronic nutritional disorder caused by long-term inadequate nutrient intake. This condition is assessed by measuring a child's length or height adjusted for age and sex. However, because of low public awareness about regularly measuring the height of children under five, stunting often goes undetected in its early stages. Stunting actually begins while the fetus is still in the womb and usually only becomes clearly visible when the child is two years old [8]. This chronic undernutrition also reduces the cognitive abilities of those affected [9].

2.1.2 Causes and Impacts of Stunting

Stunting causes one million child deaths globally every year [10]. This high mortality rate occurs because chronic undernutrition damages the immune system of young children, leaving them physically very vulnerable and unable to survive when attacked by deadly infectious diseases such as severe diarrhea and pneumonia [11]. Stunting in children under five is also triggered by various other factors, such as a history of low birth weight (LBW), births spaced too closely together, and inadequate nutrient intake. In terms of nutrition, the condition is generally caused by low consumption of fats, legumes, and diverse foods, which is often made worse by a habit of eating high-sugar foods [12], and It is also influenced by whether the child was exclusively breastfed, whether MPASI was started at the right age (around 6 months), zinc and iron intake, a history of infectious diseases, and genetic factors [13].

Stunting has very significant short-term and long-term impacts on every aspect of a child's growth and development. In the short term, chronic undernutrition causes serious delays in brain development, a significant decline in cognitive function, metabolic disorders, and a weakened immune system that makes the child prone to infectious diseases. In the long term, these negative effects continue, leading to poor academic performance at school, suboptimal adult height, and a sharply increased risk of deadly degenerative diseases such as hypertension, diabetes mellitus, and heart disease later in life. As a result, the accumulation of these physical and cognitive limitations ultimately lowers the child's work capacity and productivity once they reach working age [14]

2.2 Child Nutritional Status

Nutritional status during the golden period of development (0–5 years) is an important indicator for evaluating the health of children under five. Referring to Permenkes No. 2 of 2020, nutritional status assessment in Indonesia adopts the WHO growth standards using the *Z-score* method which measures how many standard deviations a child's anthropometric parameter lies from the median of the reference population [6]. Early detection of stunting specifically focuses on monitoring the Length- or

Height-for-Age index (LFA or HFA) as the national standard reference for accurately classifying chronic growth disorders. This theoretical concept of the Z-score, used to determine the position of a child's physical measurements relative to the population reference standard, then serves as the main computational foundation of the system's algorithm design [15].

2.3 MPASI

Complementary feeding (Makanan Pendamping ASI, or MPASI) is transitional nutritional intake given alongside breast milk, and it is important to start giving it at 6 months of age to meet the child's rising energy needs, which can no longer be met by breast milk alone [16]. Inappropriate MPASI intake, whether in terms of quantity, nutritional quality, or texture, is a major trigger of chronic malnutrition that leads to stunting and permanent cognitive developmental delays. Therefore, MPASI feeding standards must follow four main principles: timed (given when the child starts needing additional nutrients), safe (hygienic from preparation to serving), adequate (a balanced supply of nutrients), and properly scheduled (a regular and responsive feeding pattern that encourages the child's independence) [17].

2.4 Immunization

Babies are born with passive immunity transferred from the mother through the placenta and breast milk [18]. Although this natural protection is vital in the early stages of life, it is relatively short-lived and starts to decline significantly when the child reaches 6 to 12 months of age. As maternal antibodies decrease while the child's own immune system is still immature, the risk of severe infections and dangerous complications becomes very high. For this reason, immunization is a crucial step in building the child's immune system, as it safely stimulates the formation of antibodies and immunological memory.

Based on IDAI (Indonesian Pediatric Society) recommendations, basic immunization for children aged 0–12 months is given in stages, starting with Hepatitis B (<24 hours after birth), BCG, and oral Polio (bOPV) before 1 month of age. This is followed by a series of combination DTP-HB-Hib vaccines, Polio (bOPV and IPV), PCV, and Rotavirus at 2, 3, and 4 months of age, and then completed at 6–9 months with Influenza, the third PCV dose, IPV-2, the MR vaccine, and Japanese Encephalitis, before moving on to the *booster* schedule at later ages [19].

CHAPTER 3
Research Method

3.1 Research Stages

This development research produced the KawalTumbuh Android application through five sequential stages: problem identification, reference data collection, system design, implementation, and testing. The problem identification stage was carried out through a literature review of national stunting prevalence, child anthropometry regulations, and similar applications, which was then formulated into four main functional requirements: early detection of growth status based on Z-scores, pregnancy nutrition support, immunization schedule monitoring with automatic reminders, and MPASI menu recommendations based on the child's nutritional status and the family budget.

3.2 Software Development Method

The KawalTumbuh application was developed using the prototyping model, meaning it was built in stages and refined repeatedly until it met users' needs. This model was chosen because user needs in the child nutrition domain are hard to define completely from the start, so the design needs to be tested and improved repeatedly as understanding grows [20].

3.3 System Architecture

The application targets Android 7.0 (API 24) as the minimum version and API 35 as the target version, so it can run on a wide range of devices. The application is designed to be offline-first, with all calculations of nutritional status, nutrient targets, and menu ranking executed locally on the device, without a backend or network connection. This decision has three consequences: (a) the child's anthropometric data never leaves the device, so privacy is protected, (b) the application remains fully functional without internet, and (c) there are no server operating costs.

3.4 Calculation of Child Age and Gestational Age

Time calculations are based on the accumulated difference in days. The child's age (in months) is calculated from the number of days since birth divided by an average of 30.44 days per month, rounded down; gestational age (in weeks) is calculated from the number of days since the first day of the last menstrual period (HPHT) divided by seven, with trimesters divided at ≤ 13 weeks (Trimester I), ≤ 27 weeks (Trimester II), and the remainder (Trimester III); while the Estimated Due Date (HPL) is determined using Naegele's rule: HPHT plus 280 calendar days [21].

3.5 Nutritional Status Determination Method

3.5.1 Z-score Calculation with the LMS Method

A child's growth status is determined based on WHO growth standards using the LMS (Lambda-Mu-Sigma) method. The Z-score value is computed depending on the value of parameter L as follows:

$$Z = \frac{\left(\frac{X}{M}\right)^{L} - 1}{L \times S} \quad \text{if } L \neq 0, \qquad Z = \frac{\ln \frac{X}{M}}{S} \text{ if } L = 0$$

In the formula above, X represents the measured height (cm) or weight (kg) of the child, while L, M, and S are constants from the WHO reference table adjusted for the child's sex and age. If the child's age falls between two rows of the table, the parameters are linearly interpolated:

$$P = P_1 + (P_2 - P_1)\ x\ \left(\frac{u - a_1}{a_2 - a_1}\right)$$

In this equation, P is the parameter being sought, u is the child's age, a_1 and a_2 are the nearest age limits in the table that bracket it, and P_1 and P_2 are the parameter values at those two limits [22].

3.5.2 Status Classification

Table 3.1 Status classification based on Z-score

Z-score Range	Category
Z < −3 SD	Severely low
−3 SD ≤ Z < −2 SD	Low
−2 SD ≤ Z ≤ +2 SD	Normal
Z > +2 SD	High

Nutritional status is determined based on Permenkes No. 2 of 2020 using two main indicators, namely Weight-for-Age (WFA) and Height-for-Age (HFA). The classification thresholds in Table 3.1 are applied uniformly to all indicators as a simplification at this stage of development [23].

3.6 Child Nutrition Calculator Algorithm

The algorithm runs in four steps: (1) receiving input data, (2) determining nutritional status, (3) calculating energy and macronutrient targets, and (4) displaying output in the form of recipes.

3.6.1 Determining the Basic Energy Target

Table 3.2 Basic energy target per age group

Age group	Energy (kcal/day)
0–5 months	550
6–11 months	800
12–47 months	1,350
48–59 months	1,400

The energy target values in Table 3.2 refer to Permenkes RI No. 28 of 2019 on Recommended Dietary Allowances (AKG) [24].

3.6.2 Macronutrient Distribution

Total energy is divided among the three macronutrients using standard proportions of 12.5% protein, 30% fat, and 57.5% carbohydrate, which are converted into grams using energy factors of 4 kcal/g for protein and carbohydrate and 9 kcal/g for fat, according to the following equation; these proportions serve as the reference for recipe ranking in Subsection 3.6.4 [24].

$$Nutrient\ Mass\ (g) = \frac{Total\ energy\ x\ proportion}{Energy\ factor}$$

3.6.3 Adjustment Based on Nutritional Status

The application adjusts nutrient targets based on the child's growth status (Table 3.1) through two rules:

a. Energy adjustment based on the WFA indicator

If the child's status is Low, the total energy target is multiplied by a factor of 1.10; if it is Severely low, it is multiplied by a factor of 1.20. This adjustment adopts the concept of catch-up growth, in which underweight children are given a higher energy target so they can catch up on their growth [25].

b. Protein adjustment based on the HFA indicator

If the child's status is Low or Severely low, the protein proportion is increased from 12.5% to 17% of total energy, while the fat proportion is kept at 30% and the carbohydrate proportion adjusts accordingly. This increase steers menu recommendations toward higher-protein recipes for children at risk of stunting.

The multiplication factors (1.10/1.20) and the 17% protein proportion in the two rules above are heuristics at this stage of development and have not yet been clinically validated. Therefore, whenever an adjustment is active, the application interface shows the reason for the increased target along with a recommendation to consult a health worker or Posyandu cadre.

3.6.4 Recipe Ranking

Recipe options are limited to the age group that matches the child, and are then ranked using one of the following two scenarios. If both of the child's nutritional status indicators (WFA and HFA) are Normal or High, recipes are sorted by how close their calories are to the energy needed per meal, which is the daily energy target divided by four. However, if one or both indicators call for catch-up growth, the system considers both the calories and the protein of each recipe, where each recipe is given a combined suitability score:

$$Score = \min\left(\frac{E_r}{E_m}, 1\right) + \min\left(\frac{p_r}{p_t}, 1\right)$$

where E_r is the energy content of the recipe per serving, E_m is the energy need per meal, p_r is the percentage of the recipe's energy that comes from protein, and p_t is the target protein percentage (12.5% or 17%, following Subsection 3.6.3). Recipes with the highest score are ranked first, and recipes with the same score are sorted using the calorie-closeness rule from the first scenario. The value of each term is deliberately capped at a maximum of 1 so that recipes which nutrients far exceed the target do not automatically rank at the top.

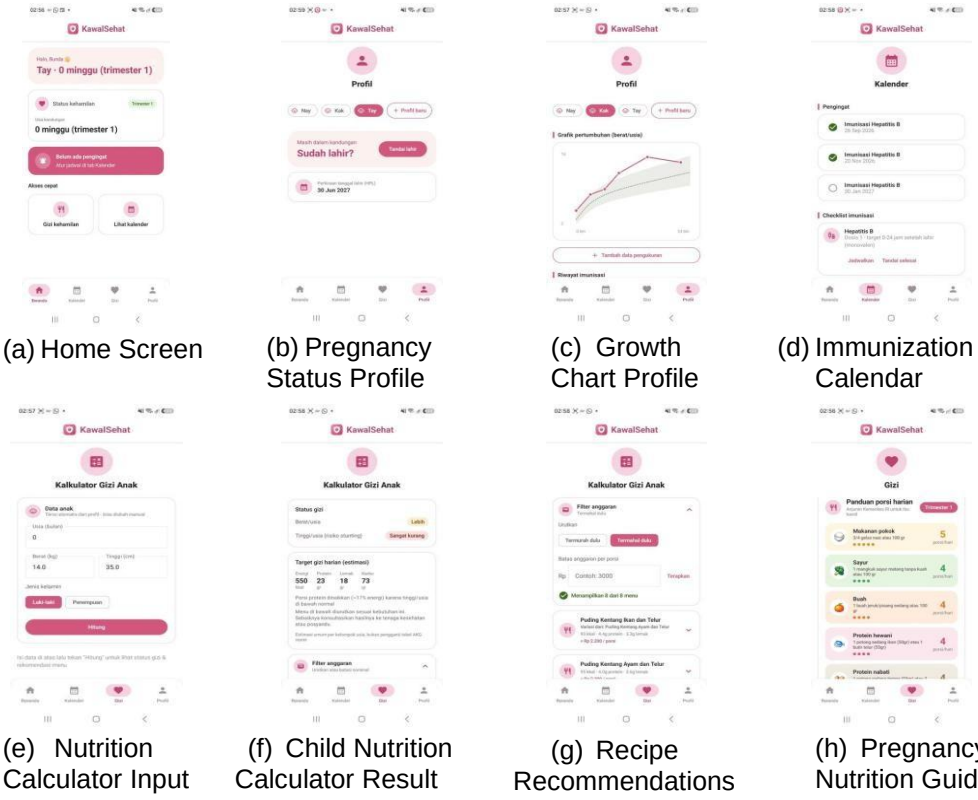
3.7 Testing Procedure

Application testing was carried out in two stages: functional testing using the black-box method on all of the application's main flows, computational verification of the Z-score formula, and the nutrition calculator algorithm by comparing them against the WHO standard tables [26]. Usability testing assessed the acceptance and benefits of the application for its main target users, namely parents of children under five, using a 1–5 Likert-scale questionnaire with four assessment aspects (Table 4.2), which was distributed to 30 parents after they tried the application directly [27]. Details of the test scenarios and results are presented in Chapter 4.

CHAPTER 4
Results and Discussion

4.1 Application Interface Implementation

The KawalTumbuh application was successfully implemented as an Android application according to the design in Chapter 3. The interface follows Material Design 3 principles with a bottom navigation bar of four main menus (Home, Calendar, Nutrition, Profile), keeping the navigation flow simple for users with varying levels of digital literacy. The eight main interfaces of the application are shown in Figure 4.1.



(a) Home Screen (b) Pregnancy Status Profile (c) Growth Chart Profile (d) Immunization Calendar

(e) Nutrition Calculator Input (f) Child Nutrition Calculator Result (g) Recipe Recommendations (h) Pregnancy Nutrition Guide

Figure 4.1 Main interfaces of the KawalTumbuh application

Home and Profile (a-c) display the user's status, active Immunization reminders, and growth charts; Calendar (d) displays the automatic immunization schedule; the Calculator (e-f) produces the Z-score status and macronutrient targets; while Nutrition (g-h) contains budget-filtered recipe recommendations sorted according to Subsection 3.6.4 as well as pregnancy nutrition guidance.

4.2 Functional Testing Results

Functional testing was carried out using the *black-box testing* method on eight main scenarios: onboarding for the pregnant mother path, onboarding for the child path, automatic transition from the pregnancy profile to the child profile after birth, entering the child's weight/height measurements, the nutrition calculator for a normal status, the nutrition calculator for a Severely low HFA status, filtering recipe recommendations

by budget, and using the application in airplane mode. All eight scenarios produced the expected results (Passed), including confirmation that all features remain fully functional without an internet connection, in line with the *offline-first* architecture designed in Chapter 3.

4.3 Z-score Calculation Verification Results

Calculation verification was carried out to ensure that the implementation of the Z-score formula with the LMS method in the application produces values consistent with the WHO anthropometric standard tables.

Table 4.1 Z-score calculation verification results

Case	Child Profile (Sex, age, weight, height)	Indicator	Application Z-score	WHO Reference Z-score	Difference
1	Male, 12 months, 9.6 kg, 75.7 cm	WFA	−0.06	−0.05	0.01
		HFA	0.00	0.00	0.00
2	Female, 18 months, 8.2 kg, 79.5 cm	WFA	−2.45	−2.44	0.01
		HFA	−1.20	−1.19	0.01
3	Male, 24 months, 10.8 kg, 80.3 cm	WFA	−1.80	−1.79	0.01
		HFA	−3.21	−3.20	0.01

The maximum difference between the Z-score values produced by the application and the WHO reference values is 0.01, which is well below the 0.1 SD tolerance threshold commonly used in anthropometric standardization. Thus, the implementation of the Z-score formula in the application can be declared computationally valid.

4.4 Usability Testing Results

Table 4.2 Summary of usability testing results

No	Assessment Aspect	Average Score (1–5)	Percentage Level	Category
1	Ease of Use	4.43	88.67%	Very Good
2	Feature Usefulness	4.70	94.00%	Very Good
3	Clarity of Information	4.67	93.33%	Very Good
4	User Satisfaction	4.80	96.00%	Very Good
	Overall Average	**4.65**	**93.00%**	**Very Good**

Based on the usability questionnaire completed by 30 respondents who are parents of children under five, the application received the highest scores for user satisfaction (4.80) and feature usefulness (4.70), while ease of use scored slightly lower (4.43). Overall, the average percentage score reached 93.00%, which according to Riduwan's score interpretation criteria falls into the Very Good category, showing that the KawalTumbuh application is well accepted by its target users [27].

CHAPTER 5

Conclusion

KawalTumbuh was successfully developed as an offline-first Android application that integrates a Z-score-based child nutrition calculator, budget-friendly MPASI recipe recommendations, immunization schedule tracking, and pregnancy nutrition guidance in a single platform. The implementation of the LMS formula was shown to be computationally valid, with a maximum difference of 0.01 SD from the WHO reference. All black-box testing scenarios passed, including without an internet connection. Because all calculations run on the device, children's anthropometric data stays protected and there are no server costs.

Based on the usability test with 30 parents of children under five, the application obtained an average score of 4.65 (93.00%), in the Very Good category. User satisfaction was the highest-scoring aspect (4.80), while ease of use was the lowest (4.43), suggesting that the interface flow could still be simplified. These results show that the application is well accepted, but they do not yet measure its effect on reducing stunting cases.

For future development, the energy multiplication factors (1.10/1.20) and the 17% protein proportion, which are still heuristics, need to be validated together with nutritionists or health workers. Furthermore, longitudinal testing with a broader participant pool is necessary, alongside an expansion of the local recipe database.

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